Explicit Gaussian polynomial-threshold pseudorandom generator with polynomial dependence on degree

Zander Kelley, Raghu Meka

doi:10.4230/LIPIcs.CCC.2022.21; CC BY 4.0

Editorial scope. Counted original Theorem3 once, the explicit PRG for Gaussian degree-d PTFs with its original polynomial-degree seed bound and parameter range. Complete author quantitative generator, Hermite/vector anticoncentration, derivative growth, mollifier and main two-case hybrid/remainder proof included. Structural/support statements and direct applications are not extra tasks. Text is from the officially published CC BY4 version, not a redistribution of unpublished nonexclusive arXiv exposition.

Allowed standard prerequisites (editorial).

- Gaussian Hermite orthogonality/hypercontractivity [13], imported Kane relative anticoncentration [9] and generator discretization [7], elementary moment and Taylor bounds. Exact imported Theorems12/Lemma18 and Gaussian basis/noise facts retained. All source-local vector anticoncentration, slow-growth, Hermite restrictions, deviation moments, mollifier and main hybrid-step proofs retained; cited background not recursively reproved.

Necessary original definitions and main result

Let $\mathrm{sign} : \mathbb{R} \to \mathbb{R}$ be defined as $\mathrm{sign}(z) = 1$ if $z \geq 0$ and 0 otherwise.

Definition 1. For an integer $d > 0$, a degree d PTF $f : \mathbb{R}^n \to \{0,1\}$ is a function of the form $f(x) = \mathrm{sign}(p(x))$, where $p : \mathbb{R}^n \to \mathbb{R}$ is a polynomial of degree at most d. Our goal is to design a PRG that takes few random bits and outputs a high-dimensional vector whose distribution is indistinguishable from a standard multivariate gaussian by any low-degree PTF. Specifically:

Definition 2. A function $G : \{0,1\}^r \to \mathbb{R}^n$ is a pseudorandom generator for degree d PTFs with error ε if for every degree at most d PTF $f : \mathbb{R}^n \to \{0,1\}$,

$$\left|\mathbb{P}_{y \in_u \{0,1\}^r}(f(G(y)) = 1) - \mathbb{P}_{x \sim N(0,1)^n}(f(x) = 1)\right| \leq \varepsilon.$$

We call r the seed length of the generator and say G ε-fools degree d PTFs with respect to the gaussian distribution.[1] We say G is explicit if its output can be computed in time polynomial in n.

Theorem 3 (PRG for PTFs). There exist constants c, C such that for all $\varepsilon > 0$ and $d \geq 1$, there exists an explicit PRG that ε-fools n-variate degree d PTFs with respect to the gaussian distribution with seed length $r(n, d, \varepsilon) = Cd^c \log(n/\varepsilon) \log(1/\varepsilon)/\varepsilon^2$.

Definition 4. Given a function $p : \mathbb{R}^n \to \mathbb{R}$ and $x \in \mathbb{R}^n$, and a restriction parameter $\lambda \in (0, 1)$, let $p_{x,\lambda} : \mathbb{R}^n \to \mathbb{R}$ be the function $p_{x,\lambda}(y) = p(\sqrt{1-\lambda}x + \sqrt{\lambda}y)$.[2] Intuitively, we can view $p_{x,\lambda}$ as a restriction where $(1-\lambda)$-fraction of the variance is already fixed. (Note that for independent $x, y \sim N(0,1)^n$, $\sqrt{1-\lambda}x + \sqrt{\lambda}y$ is distributed as $N(0,1)^n$.) For a function $p : \mathbb{R}^n \to \mathbb{R}$, let $\|\nabla^k p(x)\|^2$ denote the sum of squares of all partial derivatives of p of order k at x. That is, $\|\nabla^k p(x)\|$ is the Frobenius norm of the tensor of k'th order partial derivatives of p. We show that for any degree d polynomial p, the Frobenius-norm of the k'th order derivatives are comparable to the $(k-1)$'th order derivatives on a random gaussian input with high probability:

Lemma 6. For any degree-d polynomial $p : \mathbb{R}^d \to \mathbb{R}$, and $x \sim N(0,1)^n$, the following holds with probability at least $1 - \delta$:

$$\|\nabla^k p(x)\| \leq O(d^3/\delta)\|\nabla^{k-1} p(x)\|, \text{ for all } 1 \leq k \leq d. \tag{1}$$

Definition 10. For a polynomial $p : \mathbb{R}^n \to \mathbb{R}$ of the form $p(y) := \sum_\alpha \widehat{p}(\alpha) h_\alpha(y)$, define its hypervariance, $\mathrm{HyperVar}_R(\ )$, and normalized hypervariance, $H_R(\ )$, as

$$\mathrm{HyperVar}_R(p) := \sum_{\alpha \neq 0} \widehat{p}(\alpha)^2 R^{2|\alpha|}, \qquad H_R(p) := \frac{\mathrm{HyperVar}_R(p)}{\widehat{p}(0)^2}.$$

Intuitively, if the normalized hypervariance $H_R(p)$ of a polynomial is small for a large R, then it means that the weights of the higher-order Hermite coefficients of p have a geometric decay.

3 Preliminaries

The pseudorandom generator construction: idealization vs. discretization. Following [7] and [14], we analyze the idealized pseudorandom distribution

$$Z = \frac{1}{\sqrt{L}} \sum_{i=1}^{L} Y_i,$$

where each $Y_i \in \mathbb{R}^n$ is a k-moment-matching gaussian (that is, $\mathbb{E}[p(Y_i)] = \mathbb{E}_{x \sim N(0,1)^n}[p(x)]$ for all polynomials $p : \mathbb{R}^n \to \mathbb{R}$ of degree at most k). Suppose

[1] Here, and henceforth, $y \in_u S$ denotes a uniformly random element from a multi-set S, and $N(0,1)$ represents the standard univariate gaussian distribution of variance 1.

[2] As the value of λ will often be clear, we will often in fact just use p_x for brevity.

that, for any such Z with parameters (L, k), it is the case that Z fools degree-d PTFs with error $\varepsilon = \varepsilon(L, k, d)$. Then, it is shown in [7] how to obtain a small-seed length PRG (in the sense of Definition 2) by providing a specific instantiation and discretization of this construction.

Theorem 12 ([7], implicit in Section 6). Suppose a Z as above with parameters (L, k) fools degree d-PTFs with error $\varepsilon = \varepsilon(L, k, d)$. Then, there is an explicit, efficiently computable PRG with seed length $O(dkL \log(ndL/\varepsilon))$ that (2ε)-fools degree d PTFs.

Hermite polynomials. To argue about polynomials which are not necessarily multilinear, we need some simple facts concerning Hermite polynomials. For our purposes, Hermite polynomials are simply a convenient choice of polynomial basis which have nice properties (in particular being orthonormal) with respect to gaussian inputs. For a more detailed background on Hermite polynomials and their use for analyzing functions over gaussian space, see [13, Ch. 11]. One concrete way to define the Hermite polynomials is the following:

- For the univariate polynomials, the degree-m "Probabilist's" Hermite polynomial is the m-th coefficient of the generating function

$$e^{st - \frac{1}{2}s^2} = \sum_{m \geq 0} H_m(t) s^m.$$

- We define the degree-m univariate Hermite polynomial by the normalization

$$h_m(t) := \frac{1}{\sqrt{m!}} H_m(t).$$

- For a multi-index $\alpha \in \mathbb{N}^n$, we define the multivariate Hermite polynomial $h_\alpha : \mathbb{R}^n \to \mathbb{R}$ via the product

$$h_\alpha(x) := \prod_{i=1}^{n} h_{a_i}(x_i).$$

We record some basic properties of this particular choice of polynomial basis. The final two properties say that the Hermite basis is orthonormal with respect to correlation under the standard gaussian distribution – this is the reason for our choice of normalization.

- The set $\{h_\alpha(x) : |\alpha| \leq d\}$ is a basis for real polynomials in n variables of degree $\leq d$.
- h_0 is the constant polynomial $h_0 \equiv 1$.
- For multi-indicies $\alpha \in \{0, 1\}^n$, $h_\alpha(x)$ is simply the monomial $\prod_{i:a_i=1} x_i$.
- For $x \sim N(0, 1)^n$, and distinct multi-indices $\alpha \neq \beta$, $\mathbb{E}_x h_\alpha(x) h_\beta(x) = 0$.
- For $x \sim N(0, 1)^n$, and any multi-index α, $\mathbb{E}_x h_\alpha(x)^2 = 1$.

Gaussian noise operator. We recall the definition of the noise operator U_ρ, which here we regard as an operator on real polynomials in n variables (see [13, Ch. 11] for background and a more general viewpoint). For a polynomial $f : \mathbb{R}^n \to \mathbb{R}$ and a parameter $\rho \in [0, 1]$, the action of U_ρ on f is specified by

$$(U_\rho f)(x) := \mathbb{E}_{Z \sim N(0,1)^n} f(\rho x + \sqrt{1-\rho^2} Z).$$

An important feature of the Hermite basis is that the noise operator acts on it diagonally (see [13, Ch. 11]): $U_\rho h_\alpha(x) = \rho^{|\alpha|} h_\alpha(x)$. Thus, if f is a degree-d polynomial given in the Hermite basis as

$$f(x) = \sum_{|\alpha| \le d} \widehat{f}(\alpha) h_\alpha(x),$$

then we can express the result of the noise operator applied to f explicitly as

$$U_\rho f(x) = \sum_{|\alpha| \le d} \widehat{f}(\alpha) \rho^{|\alpha|} h_\alpha(x).$$

Higher moments and hypercontractivity. Fix a polynomial $f(x) := \sum_{|\alpha| \le d} \widehat{f}(\alpha) h_\alpha(x)$. For an even natural number $q \ge 2$, we write the gaussian q-norm of f as

$$\|f\|_q := \left(\mathbb{E}_{x \sim N(0,1)^n} f(x)^q\right)^{1/q}.$$

We wish to be able to bound this quantity in terms of the magnitudes of the Hermite coefficients of f, $\widehat{f}(\alpha)$. For this purpose, we extend the definition of U_ρ also to $\rho > 1$ by its action on the Hermite basis: $U_\rho h_\alpha(x) = \rho^{|\alpha|} h_\alpha(x)$. With this notation, we can express the well-known $(q, 2)$-hypercontractive inequality [13, Ch. 9,11] as

$$\|f\|_q \le \|U_{\sqrt{q-1}} f\|_2,$$

which is quite convenient for us, as we can use orthonormality of the Hermite basis to explicitly compute

$$\|U_{\sqrt{q-1}} f\|_2^2 = \sum_{|\alpha| \le d} (q-1)^{|\alpha|} \widehat{f}(\alpha)^2 \le \sum_{|\alpha| \le d} q^{|\alpha|} \widehat{f}(\alpha)^2.$$

Gaussian restrictions and derivatives on the Hermite basis. Besides the effect of the noise operator, it will also be important to understand the effect of two further operations on polynomials:

- The derivative map, $f(y) \mapsto \partial^\alpha f(y)$.
- The gaussian restriction at x, $f(y) \mapsto f(\sqrt{1-\lambda} x + \sqrt{\lambda} y)$.

In particular, we are concerned with how these operations affect the Hermite coefficients of a polynomial; ultimately, our goal will be to develop a "Hermite-basis analogue" of the Taylor expansion which can be applied to expand $f(\sqrt{1-\lambda} x + \sqrt{\lambda} y)$ as a function of y. We start by computing the effect of these two operations on univariate Hermite polynomials, and then on the full multivariate Hermite basis, and finally on a general polynomial $f(x)$ expressed in the Hermite basis.

Proposition 14. For univariate Hermite polynomials, we have the identities

- $\frac{\partial^k}{\partial t^k} h_m(t) = \sqrt{\frac{m!}{(m-k)!}} h_{m-k}(t)$,
- $h_m(\sqrt{1-\lambda}x + \sqrt{\lambda}y) = \sum_{k=0}^m \sqrt{\binom{m}{k}} (1-\lambda)^{(m-k)/2} \lambda^{k/2} h_{m-k}(x) h_k(y)$.

Proof. The first of these identities is standard (see e.g. [13, Ex. 11.10]); we provide a proof of the second. The second identity can be proved by considering the generating function

$$e^{st - \frac{1}{2}s^2} = \sum_m \sqrt{m!} h_m(t) s^m,$$

and comparing the coefficient of s^m on both sides of

$$e^{s(\sqrt{1-\lambda}x + \sqrt{\lambda}y) - \frac{1}{2}s^2} = e^{(s\sqrt{1-\lambda})x - \frac{1}{2}(s\sqrt{1-\lambda})^2} \cdot e^{(s\sqrt{\lambda})y - \frac{1}{2}(s\sqrt{\lambda})^2}.$$

◁ The corresponding identities for multivariate Hermite polynomials follow easily from above.

Proposition 15. We have

- $\partial^\alpha h_\beta(y) = \sqrt{\alpha!/\gamma!} h_\gamma(y)$, where $\gamma = \beta - \alpha$,
- $h_\beta(\sqrt{1-\lambda}x + \sqrt{\lambda}y) = (1-\lambda)^{|\beta|/2} \sum_{\alpha \le \beta} \frac{\partial^\alpha h_\beta(x)}{\sqrt{\alpha!}} \left(\frac{\lambda}{1-\lambda}\right)^{|\alpha|/2} h_\alpha(y)$,
- $\partial^\alpha h_\beta(\sqrt{1-\lambda}x + \sqrt{\lambda}y) = (1-\lambda)^{|\beta-\alpha|/2} \sum_{\gamma \le \beta-\alpha} \frac{\partial^{\alpha+\gamma} h_\beta(x)}{\sqrt{\gamma!}} \left(\frac{\lambda}{1-\lambda}\right)^{|\gamma|/2} h_\gamma(y)$.

We conclude with a Taylor-like expansion in the Hermite basis that we use repeatedly.

Lemma 16. Let $f(y) = \sum_\alpha \widehat{f}(\alpha) h_\alpha(y)$. Then

$$f(\sqrt{1-\lambda}x + \sqrt{\lambda}y) = \sum_\alpha \frac{\partial^\alpha g(x)}{\sqrt{\alpha!}} \left(\frac{\lambda}{1-\lambda}\right)^{|\alpha|/2} h_\alpha(y),$$

where $g(x) := U_{\sqrt{1-\lambda}} f(x) = \sum_\alpha \widehat{f}(\alpha)(1-\lambda)^{|\alpha|/2} h_\alpha(x)$.

Proof. We express

$$\begin{aligned} f(\sqrt{1-\lambda}x + \sqrt{\lambda}y) &= \sum_\alpha \widehat{f}(\alpha) h_\alpha(\sqrt{1-\lambda}x + \sqrt{\lambda}y) \\ &= \sum_\alpha \frac{h_\alpha(y)}{\sqrt{\alpha!}} \left(\frac{\lambda}{1-\lambda}\right)^{|\alpha|/2} \sum_{\beta \ge \alpha} \widehat{f}(\beta)(1-\lambda)^{|\beta|/2} \partial^\alpha h_\beta(x) \\ &= \sum_\alpha \frac{h_\alpha(y)}{\sqrt{\alpha!}} \left(\frac{\lambda}{1-\lambda}\right)^{|\alpha|/2} \partial^\alpha g(x). \end{aligned}$$

◁ Lastly, we will also need an extension of this theorem which expresses $\partial^\alpha f$, at the point $\sqrt{1-\lambda}x + \sqrt{\lambda}y$, as a polynomial in y in the Hermite basis.

Theorem 17. Let $f(y) = \sum_\alpha \widehat{f}(\alpha) h_\alpha(y)$. Then

$$\partial^\alpha f(\sqrt{1-\lambda}x + \sqrt{\lambda}y) = (1-\lambda)^{-|\alpha|/2} \sum_{\beta \geq \alpha} \partial^\beta g(x) \sqrt{\frac{\alpha!}{\beta!}} \left(\frac{\lambda}{1-\lambda}\right)^{|\beta-\alpha|/2} h_{\beta-\alpha}(y),$$

where $g(x) := U_{\sqrt{1-\lambda}} f(x)$.

Proof. We express

$$\begin{aligned}
\partial^\alpha f(\sqrt{1-\lambda}x + \sqrt{\lambda}y) &= \sum_\beta \widehat{f}(\beta) \partial^\alpha h_\beta(\sqrt{1-\lambda}x + \sqrt{\lambda}y) \\
&= \sum_\gamma \frac{h_\gamma(y)}{\sqrt{\gamma!}} \left(\frac{\lambda}{1-\lambda}\right)^{|\gamma|/2} \sum_{\beta \geq \gamma+\alpha} \widehat{f}(\beta)(1-\lambda)^{|\beta-\alpha|/2} \partial^{\alpha+\gamma} h_\beta(x) \\
&= (1-\lambda)^{-|\alpha|/2} \sum_\gamma \frac{h_\gamma(y)}{\sqrt{\gamma!}} \left(\frac{\lambda}{1-\lambda}\right)^{|\gamma|/2} \partial^{\alpha+\gamma} g(x).
\end{aligned}$$

◁

4.1 Proof of Lemma 6

Our main tool will be Kane's relative-anticoncentration lemma for gaussian polynomials.

Lemma 18 ([9]). For a degree d polynomial p, and independent standard gaussian vectors $x, y \in \mathbb{R}^n$,

$$\mathbb{P}(|p(x)| \leq \varepsilon |\langle y, \nabla p(x) \rangle|) \leq O(\varepsilon d^2).$$

In fact, we will actually work with the following corollary which is essentially the first of the d inequalities in Lemma 6.

Corollary 19. For a degree d polynomial p, and independent standard gaussian vector $x \in \mathbb{R}^n$,

$$\mathbb{P}(|p(x)| \leq \varepsilon \|\nabla p(x)\|) \leq O(\varepsilon d^2).$$

Proof. We note that for any fixed x, $\langle y, \nabla p(x) \rangle$ is identical in distribution to $Z\|\nabla p(x)\|$, where $Z \sim N(0,1)$ is a standard gaussian. So, we express

$$\begin{aligned}
\mathbb{P}(|p(x)| \leq \varepsilon |\langle y, \nabla p(x) \rangle|) &= \mathbb{P}(|p(x)| \leq \varepsilon |Z| \|\nabla p(x)\|) \\
&\geq \mathbb{P}(|p(x)| \leq \varepsilon \|\nabla p(x)\|) \cdot \mathbb{P}(|Z| \geq 1).
\end{aligned}$$

Since $\mathbb{P}(|Z| \geq 1) \geq \Omega(1)$, we conclude that

$$\mathbb{P}(|p(x)| \leq \varepsilon \|\nabla p(x)\|) \leq O(\varepsilon d^2).$$

◁ The heart of the proof of Lemma 6 is a vector-valued variant of the above corollary:

Lemma 20. Let $\vec{f}(x) := (f_1(x), f_2(x), \ldots, f_m(x))$ be a collection of m degree-at-most d polynomials $f_j(x)$. If $x \in \mathbb{R}^n$ is a standard gaussian vector, then

$$\mathbb{P}\left(\|\vec{f}(x)\|^2 \leq \varepsilon^2 \sum_{j=1}^{m} \|\nabla f_j(x)\|^2 \right) \leq O(\varepsilon d^2).$$

Proof of Lemma 6. We simply apply the above lemma d times and take a union bound. For $1 \leq k \leq d$, let $\vec{f_k}(x) := ((\partial^\alpha f(x) : |\alpha| = k))$. Note that $\|\vec{f_k}(x)\|^2 = \|\nabla^k f(x)\|^2$. Further, note that

$$\sum_{\alpha:|\alpha|=k} \|\nabla(\partial^\alpha f(x))\|^2 \geq \|\nabla^{k+1} f(x)\|^2,$$

where the inequality follows as each $(k+1)$'th order derivative would be counted at least once in the expression on the left hand side. Therefore, by the above lemma, for $x \sim N(0,1)^n$, we have

$$\mathbb{P}(\|\nabla^k f(x)\|^2 \leq \varepsilon^2 \|\nabla^{k+1} f(x)\|^2) \leq O(\varepsilon d^2).$$

Setting $\varepsilon = \delta/d^3$, and taking a union bound over all k, we get that for a constant $C > 0$,

$$\mathbb{P}(\forall k, \|\nabla^k f(x)\|^2 > C(\delta^2/d^6)\|\nabla^{k+1} f(x)\|^2) \geq 1 - \delta.$$

This proves Lemma 6. ◁

Proof of Lemma 20. Consider the auxiliary polynomial

$$h(x, y) := \sum_{j=1}^{m} f_j(x) y_j.$$

As a function of both x and y, we have

$$\nabla h(x, y) = \vec{f}(x) \circ M_x y,$$

where M_x is the matrix with columns $\nabla f_j(x)$ (that is, M_x has (i, j)-th entry $\frac{\partial}{\partial x_i} f_j(x)$). So, applying Corollary 19 to this auxiliary polynomial gives the probability bound

$$\begin{aligned} q &:= \mathbb{P}(h(x,y)^2 \leq \varepsilon^2 \|\nabla g(x,y)\|^2) \\ &= \mathbb{P}(\langle y, \vec{f}(x)\rangle^2 \leq \varepsilon^2(\|\vec{f}(x)\|^2 + \|M_x y\|^2)) \\ &\leq O(\varepsilon d^2). \end{aligned}$$

Now, for some constant $C \geq 2$ to be specified later, let E denote the event that

$$(C^2 - 1)\|\vec{f}(x)\|^2 \leq \frac{\varepsilon^2}{2}\|M_x\|_F^2,$$

where $\|M_x\|_F$ is the Frobenius norm of M_x. We note that we can lower-bound the probability q by

$$q \geq \mathbb{P}(E) \cdot \mathbb{P}(\langle y, \vec{f}(x)\rangle \leq C\|\vec{f}(x)\| \text{ and } \|M_x y\|^2 \geq \tfrac{1}{2}\|M_x\|_F^2 \mid E).$$

We claim that for large enough choice of constant C, this conditional probability can be lower-bounded by $\Omega(1)$. Indeed, we can argue for any fixed x:

- $\mathbb{P}(\langle y, \vec{f}(x)\rangle \geq C\|\vec{f}(x)\|) \leq 1/C^2$.
- $\mathbb{P}(\|M_x y\|^2 \geq \frac{1}{2}\|M_x\|_F^2) \geq \Omega(1)$.

The first item is just a Chebyshev inequality; the second item can be derived e.g. from the basic anticoncentration bound one obtains for degree-2 polynomials from the Paley-Zygmund bound together with hypercontractivity (since, for any fixed matrix M, the quadratic form $g(y) := \|My\|^2$ has second-moment $\mathbb{E}g(y)^2 \geq (\mathbb{E}g(y))^2 = \|M\|_F^2$). Thus, by choosing C large enough, we can lower-bound this conditional probability by $\Omega(1) - 1/C^2 \geq \Omega(1)$. We conclude that $\mathbb{P}(E) \leq O(q) = O(\varepsilon d^2)$. This gives the desired conclusion

$$\mathbb{P}(\|\vec{f}(x)\| \leq \Omega(\varepsilon)\|M_x\|_F) \leq O(\varepsilon d^2).$$

◁

5 Pseudorandom Generator for PTFs

The following theorem gives quantitative bounds on the error of our main generator:

Theorem 21. Fix some parameters $\varepsilon > 0$ and $R \in \mathbb{N}$. Let z be a standard gaussian, and let $Z = \frac{1}{\sqrt{L}}\sum_{i=1}^{L} Y_i$, where each Y_i is dR-moment-matching. Then for some sufficiently large absolute constant c and any polynomial p of degree d,

$$\mathbb{E}_Z \operatorname{sign}(p(Z)) \geq \mathbb{E}_{z\sim N(0,1)^n} \operatorname{sign}(p(z)) - O(\varepsilon d^3) - L \cdot 2^{-\Omega(R)},$$

as long as L is at least Rd^c/ε^2. Combining the above with Theorem 12 immediately implies our main result Theorem 3.

Proof of Theorem 3. Given a target error ε', set $\varepsilon = \varepsilon'/Cd^3$, and $R = C\log(d/\varepsilon)$ for a sufficiently big constant so that the error in the above lemma is at most $\varepsilon'/2$ for $L = Rd^c/\varepsilon^2 = O(d^c \log(d/\varepsilon)/\varepsilon^2)$. While the above theorem only gives a lower bound, we can get an upper bound by applying the result to $-p$. Now, by applying Theorem 12 there exists an efficient PRG that fools degree d PTFs with error at most ε' and seed length $O(d^{O(1)}\log(nd/\varepsilon')\log(d/\varepsilon')/(\varepsilon')^2)$ which can be simplified to the bound in the theorem. ◁

We now prove the above theorem by the lower-sandwiching argument outlined in Section 2.1. Fix a polynomial $p(x)$ of degree d. We remind the reader of our convention $\operatorname{sign}(t) := \mathbf{1}(t \geq 0)$. We define the mollifier function

$$g(x) := \prod_{k=0}^{d-1} \rho\left(\log\left(\frac{1}{16\varepsilon^2}\frac{\|\nabla^k p(x)\|^2}{\|\nabla^{k+1}p(x)\|^2}\right)\right),$$

where $\rho : \mathbb{R} \to [0,1]$ is some smooth univariate function with $\rho(t) = 0$ for $t \leq 0$, $\rho(t) = 1$ for $t \geq 1$, and $\|\frac{\partial^k \rho}{\partial t^k}\|_\infty \leq k^{O(k)}$ for all k.[3]

[3]For example, it suffices to let $\rho(t)$ be the standard mollifier $\rho(t) := 0$ for $t \leq 0$, $\rho(t) := 1$ for $t \geq 1$, and $\rho(t) := e \cdot \exp(1/((t-1)^2 - 1))$ for $t \in (0,1)$.

Proof of Theorem 21. For every point $x \in \mathbb{R}^n$ we have

$$\operatorname{sign}(p(x)) \geq \operatorname{sign}(p(x))g(x).$$

Furthermore, under the truly-random gaussian inputs $z \sim N(0,1)^n$ we have

$$\mathbb{E}_z \operatorname{sign}(p(z))g(z) \geq \mathbb{E}_z \operatorname{sign}(p(z)) - \mathbb{E}_z|g(z)-1| \geq \mathbb{E}_z \operatorname{sign}(p(z)) - O(\varepsilon d^3),$$

where the final inequality here follows from Lemma 6. Combining these, we get that

$$\begin{aligned}\mathbb{E}_Z \operatorname{sign}(p(Z)) \geq \mathbb{E}_{z\sim N(0,1)^n} \operatorname{sign}(p(z)) - O(\varepsilon d^3) \\ - \left|\mathbb{E}_Z \operatorname{sign}(p(Z))g(Z) - \mathbb{E}_{z\sim N(0,1)^n} \operatorname{sign}(p(z))g(z)\right|.\end{aligned}$$

Thus, it suffices to bound $|\mathbb{E}_Z \operatorname{sign}(p(Z))g(Z) - \mathbb{E}_{z\sim N(0,1)^n} \operatorname{sign}(p(z))g(z)|$, which we do by a hybrid argument. We first represent z as $z := \frac{1}{\sqrt{L}}\sum_{i=1}^{L} y_i$ where each y_i is an independent standard gaussian. We can replace each Y_i with y_i and get

$$|\mathbb{E}_Z \operatorname{sign}(p(Z))g(Z) - \mathbb{E}_y \operatorname{sign}(p(y))g(y)| \leq 2^{-\Omega(R)}L,$$

as a consequence of the following lemma (restatement of Lemma 8) that we prove in the next section. Theorem 21 now follows. ◁

Lemma 22 (Main hybrid-step). There exists a constant c such that the following holds for $\lambda \leq \varepsilon^2/Rd^c$. For any fixed vector $x \in \mathbb{R}^n$, Y a dR-moment-matching gaussian vector, and $y \sim N(0,1)^n$,

$$|\mathbb{E}_Y \operatorname{sign}(p(x+\sqrt{\lambda}Y))g(x+\sqrt{\lambda}Y) - \mathbb{E}_y \operatorname{sign}(p(x+\sqrt{\lambda}y))g(x+\sqrt{\lambda}y)| \leq \gamma = 2^{-\Omega(R)}.$$

5.1 Analysis of the main hybrid-step

The proof of Lemma 22 is by a case-analysis as outlined in the introduction. Consider the setting as in the lemma and define

$$\varphi(z) := U_{\sqrt{1-\lambda}}p\left(\frac{z}{\sqrt{1-\lambda}}\right).$$

The core argument will be a case-analysis on the derivatives of φ at the fixed point x and whether these are slow-growing. Note that if p were multi-linear, then we would simply have $\varphi \equiv p$. The starting point is the following re-scaling of Lemma 16:

$$p(x+\sqrt{\lambda}y) = \sum_{|\alpha|\leq d} \frac{\partial^\alpha \varphi(x)}{\sqrt{\alpha!}} \lambda^{|\alpha|/2} h_\alpha(y). \tag{4}$$

Further, by a re-scaling of Theorem 17, we get the following identity which gives a nice nearly self-referential expression relating the derivatives of p to those of φ:

$$\partial^\alpha p(x+\sqrt{\lambda}y) = \sum_{\beta\geq\alpha} \sqrt{\frac{\alpha!}{\beta!}} \partial^\beta \varphi(x) \lambda^{|\beta-\alpha|/2} h_{\beta-\alpha}(y). \tag{5}$$

Now, note that for a truly random gaussian y we have $\partial^\alpha\varphi(x) = \mathbb{E}_y \partial^\alpha p(x+\sqrt{\lambda}y)$. Thus, it is reasonable to expect that for typical points x and small enough

λ, $\partial^\alpha p(x+\sqrt{\lambda}y)$ will be strongly concentrated around $\partial^\alpha \varphi(x)$. The following lemma gives quantitative bounds on how much the derivatives $\partial^\alpha p(x+\sqrt{\lambda}y)$ deviate from their expectations $\partial^\alpha\varphi(x)$ for a random $y \sim N(0,1)^n$. As we will need such bounds even for k-moment-matching Y, we state the deviation bound in terms of moments:

Lemma 23. Suppose f is a degree-d polynomial, and let $\varphi(z) = U_{\sqrt{1-\lambda}} f(z/\sqrt{1-\lambda})$. Consider the polynomial

$$D(y) := \|\nabla^k f(x+\sqrt{\lambda}y) - \nabla^k \varphi(x)\|^2,$$

which measures the euclidean distance between the k-th order derivatives $\nabla^k f(x+\sqrt{\lambda}y)$ and their expectations $\nabla^k\varphi(x)$. For $y \sim N(0,1)^n$, we have the moment bound

$$\|D(y)\|_{q/2} \le \sum_{t=k+1}^{d} (\lambda d q)^{t-k} \|\nabla^t \varphi(x)\|^2.$$

That is,

$$\left(\mathbb{E}_{y\sim N(0,1)^n} \|\nabla^k f(x+\sqrt{\lambda}y) - \nabla^k\varphi(x)\|^q\right)^{1/q} \le \sqrt{\sum_{t=k+1}^{d} (\lambda d q)^{t-k} \|\nabla^t\varphi(x)\|^2}.$$

Proof. We express

$$\begin{aligned} D(y) &= \sum_\alpha (\partial^\alpha f(x+\sqrt{\lambda}y) - \partial^\alpha\varphi(x))^2 \\ &= \sum_\alpha \left(\sum_{\beta>\alpha} \sqrt{\frac{\alpha!}{\beta!}} \partial^\beta\varphi(x) \lambda^{|\beta-\alpha|/2} h_{\beta-\alpha}(y) \right)^2. \end{aligned}$$

First, by triangle-inequality, we get

$$\begin{aligned} \|D(y)\|_{q/2} &\le \sum_\alpha \left\| \left(\sum_{\beta>\alpha} \sqrt{\frac{\alpha!}{\beta!}} \partial^\beta\varphi(x) \lambda^{|\beta-\alpha|/2} h_{\beta-\alpha}(y) \right)^2 \right\|_{q/2} \\ &= \sum_\alpha \left\| \sum_{\beta>\alpha} \sqrt{\frac{\alpha!}{\beta!}} \partial^\beta\varphi(x) \lambda^{|\beta-\alpha|/2} h_{\beta-\alpha}(y) \right\|_q^2. \end{aligned}$$

Applying hypercontractivity, we now get

$$\begin{aligned} \|D(y)\|_{q/2} &\le \sum_\alpha \left\| U_{\sqrt{q}} \sum_{\beta>\alpha} \sqrt{\frac{\alpha!}{\beta!}} \partial^\beta\varphi(x) \lambda^{|\beta-\alpha|/2} h_{\beta-\alpha}(y) \right\|_2^2 \\ &= \sum_\alpha \sum_{\beta>\alpha} \frac{\alpha!}{\beta!} \partial^\beta\varphi(x)^2 \lambda^{|\beta-\alpha|} q^{|\beta-\alpha|} \\ &\le \sum_\alpha \sum_{\beta>\alpha} \partial^\beta\varphi(x)^2 \lambda^{|\beta-\alpha|} q^{|\beta-\alpha|} \end{aligned}$$

$$= \sum_{t=k+1}^{d} \binom{t}{t-k} (\lambda q)^{t-k} \|\nabla^t \varphi(x)\|^2$$
$$\leq \sum_{t=k+1}^{d} (\lambda d q)^{t-k} \|\nabla^t \varphi(x)\|^2.$$

◁ We are now ready to prove Lemma 22.

Proof of Lemma 22. We study two cases:

1. x is poorly-behaved for φ. In this case, we will show that $g(x+\sqrt{\lambda}Y) = 0$ with probability at least $1 - 2^{-\Omega(R)}$.
2. x is well-behaved for φ: In this case, we will exploit the fact that $\text{sign}(p(x+\sqrt{\lambda}Y))$ will equal $\text{sign}(\varphi(x))$ with probability $1-2^{-\Omega(R)}$. We then have to show that Y fools the mollifier g which is a bit technically involved (hence we deal with this case second unlike in Section 2.1).

We begin with the first case.

Case 1: x is poorly-behaved for φ. Consider the case where the inequality $\|\nabla^k \varphi(x)\| \geq \varepsilon \|\nabla^{k+1}\varphi(x)\|$ is violated for some k, and indeed let k_0 be the largest k such that this inequality is violated. We will argue that with probability at least $1 - 2^{-\Omega(R)}$, over random choice of Y, that

$$\|\nabla^{k_0} p(x + \sqrt{\lambda}Y)\| \leq 4\varepsilon \|\nabla^{k_0+1} p(x + \sqrt{\lambda}Y)\|,$$

in which case $g(x + \sqrt{\lambda}Y) = 0$. More specifically, we will show that it is highly likely that both

- $\|\nabla^{k_0} p(x + \sqrt{\lambda}Y)\| \leq 2\varepsilon \|\nabla^{k_0+1}\varphi(x)\|$, and
- $\|\nabla^{k_0+1} p(x + \sqrt{\lambda}Y)\| \geq \frac{1}{2}\|\nabla^{k_0+1}\varphi(x)\|$.

For this, we will use Equation (5) and Lemma 23. Supposing k_0 is the largest k such that $\|\nabla^k \varphi(x)\| < \varepsilon \|\nabla^{k+1}\varphi(x)\|$, we have

- $\|\nabla^{k_0}\varphi(x)\| \leq \varepsilon \|\nabla^{k_0+1}\varphi(x)\|$ and
- $\|\nabla^{k_0+1}\varphi(x)\| \geq \varepsilon^t \|\nabla^{k_0+1+t}\varphi(x)\|$ for all $t \geq 0$.

Lemma 23 therefore gives the bounds

$$\left(\mathbb{E}_Y \|\nabla^{k_0} p(x + \sqrt{\lambda}Y) - \nabla^{k_0}\varphi(x)\|^R\right)^{1/R} \leq \varepsilon \|\nabla^{k_0+1}\varphi(x)\| \sqrt{\sum_{t\geq 1} (\lambda d R/\varepsilon^2)^t}$$

and

$$\left(\mathbb{E}_Y \|\nabla^{k_0+1} p(x + \sqrt{\lambda}Y) - \nabla^{k_0+1}\varphi(x)\|^R\right)^{1/R} \leq \|\nabla^{k_0+1}\varphi(x)\| \sqrt{\sum_{t\geq 1} (\lambda d R/\varepsilon^2)^t}.$$

So, as long as $\lambda d R/\varepsilon^2$ is at most a sufficiently small constant, we conclude that the following bounds hold with probability at least $1 - 2^{-R}$:

$$\|\nabla^{k_0} p(x + \sqrt{\lambda}Y)\| \leq \|\nabla^{k_0}\varphi(x)\| + \|\nabla^{k_0} p(x + \sqrt{\lambda}Y) - \nabla^{k_0}\varphi(x)\|$$

$$\leq 2\varepsilon\|\nabla^{k_0+1}\varphi(x)\|, \text{ and}$$
$$\|\nabla^{k_0+1}p(x+\sqrt{\lambda}Y)\| \geq \|\nabla^{k_0+1}\varphi(x)\| - \|\nabla^{k_0+1}p(x+\sqrt{\lambda}Y) - \nabla^{k_0+1}\varphi(x)\|$$
$$\geq \tfrac{1}{2}\|\nabla^{k_0+1}\varphi(x)\|.$$

In the case that these bounds hold, we get

$$\|\nabla^{k_0}p(x+\sqrt{\lambda}Y)\| \leq 4\varepsilon\|\nabla^{k_0+1}p(x+\sqrt{\lambda}Y)\|,$$

and so $g(x+\sqrt{\lambda}Y) = 0$. As this holds with probability at least $1 - 2^{-\Omega(R)}$ for both $y \sim N(0,1)^n$ as well as Y, the conclusion of Lemma 22 follows. This finishes the proof of Case 1. ⌟

Case 2: x is well-behaved for φ. We now consider the complimentary case where

$$\|\nabla^k\varphi(x)\| \geq \varepsilon\|\nabla^{k+1}\varphi(x)\|$$

for all $k = 0, 1, \ldots, d-1$. Consider the normalized polynomial

$$f(y) := \frac{p(x+\sqrt{\lambda}y)}{\varphi(x)} = 1 + \frac{1}{\varphi(x)}\sum_{\alpha\neq 0}\partial^\alpha\varphi(x)\lambda^{|\alpha|/2}h_\alpha(y).$$

Using hypercontractivity, we bound the R-th moment of $f(y)-1$ by its $\sqrt{R}$-hypervariance:

$$\|f(y)-1\|_R \leq \|U_{\sqrt{R}}(f(y)-1)\|_2 \leq \sqrt{\sum_{k\geq 1}(\lambda R/\varepsilon^2)^k} \leq 1/2.$$

So, by a Markov argument, we have

$$\mathbb{P}(\text{sign}(p(x+\sqrt{\lambda}Y)) \neq \text{sign}(\varphi(x))) \leq 2^{-R},$$

and this holds whenever Y is k-moment-matching for $k \geq dR$. So, $\text{sign}(p(x+\sqrt{\lambda}Y))$ is nearly a constant for random Y; it remains to show that Y fools $g(x+\sqrt{\lambda}Y)$. We do this by (essentially) truncating the Taylor-series of g about x so that we are left with a degree dR polynomial, which is fooled by Y. The truncation-error will be small because our assumption, $\|\nabla^k\varphi(x)\| \geq \varepsilon\|\nabla^{k+1}\varphi(x)\|$ for all k, gives us good control on the R-th order moments of the deviations $\|\nabla^k\varphi(x)\| - \|\nabla^k p(x+\sqrt{\lambda}Y)\|$. The exact calculations are somewhat cumbersome and are given below. We will show that Y fools the mollifier function

$$g(x+\sqrt{\lambda}y) = \prod_{k=0}^{d-1}\rho\left(\log\left(\frac{1}{16\varepsilon^2}\frac{\|\nabla^k p(x+\sqrt{\lambda}y)\|^2}{\|\nabla^{k+1}p(x+\sqrt{\lambda}y)\|^2}\right)\right).$$

To simplify notation we define the shifted function $\sigma(t) := \rho(t - \log(16\varepsilon^2))$, and express

$$g(x+\sqrt{\lambda}y) = \prod_{k=0}^{d-1}\sigma(\log\|\nabla^k p(x+\sqrt{\lambda}y)\|^2 - \log\|\nabla^{k+1}p(x+\sqrt{\lambda}y)\|^2).$$

It will be convenient to think of g (redundantly) as function of $2d$ auxiliary variables $s_1 \ldots s_d, t_1, \ldots, t_d$, which we will eventually fix to

- $s_i := \|\nabla^{i-1} p(x + \sqrt{\lambda} y)\|^2$
- $t_i := \|\nabla^i p(x + \sqrt{\lambda} y)\|^2$,

so we write $g(s,t) := \prod_{i=1}^{d} \sigma(\log(s_i) - \log(t_i))$. We Taylor-expand $g(s,t)$ around the points

- $a_i := \|\nabla^{i-1} \varphi(x)\|^2$
- $b_i := \|\nabla^i \varphi(x)\|^2$,

which gives $g(s,t) = \ell(s,t) + h(s,t)$, with low-degree part

$$\ell(s,t) := \sum_{\substack{\alpha,\beta \in \mathbb{N}^d \\ |\alpha|+|\beta|<R}} \frac{\partial_s^\alpha \partial_t^\beta g(a,b)}{\alpha!\beta!} (s-a)^\alpha (t-b)^\beta$$

and remainder

$$|h(s,t)| \leq \sum_{\substack{\alpha,\beta \in \mathbb{N}^d \\ |\alpha|+|\beta|=R}} \frac{|\partial_s^\alpha \partial_t^\beta g(s^*,t^*)|}{\alpha!\beta!} |s-a|^\alpha |t-b|^\beta,$$

where "$|\partial_s^\alpha \partial_t^\beta g(s^*,t^*)|$" is notation for the maximum magnitude of $\partial_s^\alpha \partial_t^\beta g$ on any point on the line segment from (a,b) to (s,t). We need the following fact to bound the size of the derivatives of g,

Claim 24. Suppose σ is a smooth univariate function with uniform derivative bounds $\|\sigma^{(n)}\|_\infty \leq n^{O(n)}$. The bivariate function $r(u,v) := \sigma(\log(u) - \log(v))$ has derivatives bounded in size by

$$\left| \frac{\partial^n}{\partial u^n} \frac{\partial^m}{\partial v^m} r(u,v) \right| \leq \frac{n^{O(n)} m^{O(m)}}{|u|^n |v|^m}.$$

This claim follows easily from the generalized chain rule (Faà di Bruno's formula). As a result, we get the derivative bounds

$$|\partial_s^\alpha \partial_t^\beta g(s,t)| \leq \frac{|\alpha|^{O(|\alpha|)}}{|s^\alpha|} \frac{|\beta|^{O(|\beta|)}}{|t^\beta|}.$$

Using this, we bound the remainder

$$|h(s,t)| \leq \sum_{\substack{\alpha,\beta \in \mathbb{N}^d \\ |\alpha|+|\beta|=R}} d^{O(R)} \prod_{i=1}^{d} \left(\frac{|1 - s_i/a_i|}{1 - |1 - s_i/a_i|} \right)^{\alpha_i} \left(\frac{|1 - t_i/b_i|}{1 - |1 - t_i/b_i|} \right)^{\beta_i}.$$

Now, consider the event E (which depends on y) that

$$(1-\delta)\|\nabla^i \varphi(x)\|^2 \leq \|\nabla^i p(x + \sqrt{\lambda} y)\|^2 \leq (1+\delta)\|\nabla^i \varphi(x)\|^2$$

holds for all i, where $\delta \leq 1/2$ is a parameter we will set shortly. In the case that this indeed holds, we get $|h(s,t)| \leq d^{O(R)} O(\delta)^R$. We set δ just small enough to

ensure $|h(s,t)| \le 2^{-R}$. Now, we express g (which we now think of as a function of the underlying variable y) as

$$\begin{aligned} g &= g \cdot \mathbf{1}_E + g \cdot \mathbf{1}_{\overline{E}} \\ &= \ell \cdot \mathbf{1}_E + h \cdot \mathbf{1}_E + g \cdot \mathbf{1}_{\overline{E}} \\ &= \ell - \ell \cdot \mathbf{1}_{\overline{E}} + h \cdot \mathbf{1}_E + g \cdot \mathbf{1}_{\overline{E}}, \end{aligned}$$

and we obtain the pointwise bound

$$|g - \ell| \le 2^{-R} + \mathbf{1}_{\overline{E}} + |\ell| \cdot \mathbf{1}_{\overline{E}}.$$

On average over Y, we get truncation error

$$\begin{aligned} \mathbb{E}_Y |g(x + \sqrt{\lambda}Y) - \ell(Y)| &\le 2^{-R} + \mathbb{E}_Y \mathbf{1}_{\overline{E}}(Y) + \sqrt{\mathbb{E}_Y \ell^2(Y)} \sqrt{\mathbb{E}_Y \mathbf{1}_{\overline{E}}(Y)} \\ &\le 2^{-R} + O(d/\delta)^R \cdot (\lambda d R/\varepsilon^2)^{-\Omega(R)} \\ &\le 2^{-R} + d^{O(1)} \cdot (\lambda d R/\varepsilon^2)^{-\Omega(R)} \end{aligned}$$

where the second inequality here follows from the moment bounds in Lemma 23. As required by the conditions of Lemma 22, we insist that λ is small enough that this error is at most $2^{-\Omega(R)}$. Since this bound holds also for truly-random standard gaussian y, and $\mathbb{E}_Y \ell(Y) = \mathbb{E}_y \ell(y)$, we obtain the desired bound

$$|\mathbb{E}_Y g(x + \sqrt{\lambda}Y) - \mathbb{E}_y g(x + \sqrt{\lambda}y)| \le 2^{-\Omega(R)}.$$

This finishes the proof in Case 2 and hence of Lemma 22. ◁

References

[1] Ido Ben-Eliezer, Shachar Lovett, and Ariel Yadin. Polynomial threshold functions: Structure, approximation and pseudorandomness. CoRR, abs/0911.3473, 2009. arXiv:0911.3473.

[2] Ilias Diakonikolas, Daniel M Kane, and Jelani Nelson. Bounded independence fools degree-2 threshold functions. In 2010 IEEE 51st Annual Symposium on Foundations of Computer Science, pages 11–20. IEEE, 2010.

[3] Ilias Diakonikolas, Prasad Raghavendra, Rocco A. Servedio, and Li-Yang Tan. Average sensitivity and noise sensitivity of polynomial threshold functions. SIAM J. Comput., 43(1):231– 253, 2014. doi:10.1137/110855223.

[4] Prahladh Harsha, Adam Klivans, and Raghu Meka. Bounding the sensitivity of polyno- mial threshold functions. Theory of Computing, 10(1):1–26, 2014. doi:10.4086/toc.2014.v010a001.

[5] Valentine Kabanets, Daniel M Kane, and Zhenjian Lu. A polynomial restriction lemma with applications. In Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing, pages 615–628, 2017.

[6] Daniel M Kane. k-independent gaussians fool polynomial threshold functions. In 2011 IEEE 26th Annual Conference on Computational Complexity, pages 252–261. IEEE Computer Society, 2011.

[7] Daniel M Kane. A small PRG for polynomial threshold functions of gaussians. In 2011 IEEE 52nd Annual Symposium on Foundations of Computer Science, pages 257–266. IEEE, 2011.

[8] Daniel M Kane. A structure theorem for poorly anticoncentrated gaussian chaoses and applications to the study of polynomial threshold functions. In 2012 IEEE 53rd Annual Symposium on Foundations of Computer Science, pages 91–100. IEEE, 2012.

[9] Daniel M Kane. The correct exponent for the Gotsman-Linial conjecture. In 2013 IEEE Conference on Computational Complexity, pages 56–64. IEEE, 2013.

[10] Daniel M Kane. A pseudorandom generator for polynomial threshold functions of gaussian with subpolynomial seed length. In 2014 IEEE 29th Conference on Computational Complexity (CCC), pages 217–228. IEEE, 2014.

[11] Daniel M Kane. A polylogarithmic PRG for degree 2 threshold functions in the gaussian setting. In Proceedings of the 30th Conference on Computational Complexity, pages 567–581, 2015.

[12] Raghu Meka and David Zuckerman. Pseudorandom generators for polynomial threshold functions. SIAM Journal on Computing, 42(3):1275–1301, 2013.

[13] Ryan O'Donnell. Analysis of boolean functions. Cambridge University Press, 2014.

[14] Ryan O'Donnell, Rocco A Servedio, and Li-Yang Tan. Fooling gaussian PTFs via local hyperconcentration. In Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing, pages 1170–1183, 2020.

[15] Ryan O'Donnell, Rocco A. Servedio, Li-Yang Tan, and Daniel Kane. Fooling gaussian PTFs via local hyperconcentration, 2021. arXiv:2103.07809. arXiv:2103.07809.